

Data Management Project

Performance Evaluation System (PES)

An Intelligent Academic Management Platform with AI-Powered Insights

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Table of Contents

1. Introduction
2. Problem Statement & Objectives
3. Literature Review & Existing Systems
4. Novelty & Unique Features
5. Dataset Overview
6. System Architecture
7. AI Model – Training & Accuracy
8. Tools & Technologies
9. Implementation Plan & Timeline
10. Expected outcomes
11. Conclusion
12. References

1. Introduction

The **Performance Evaluation System (PES)** is a comprehensive web-based platform designed to transform how undergraduate students at the Faculty of Engineering, University of Sri Jayewardenepura (USJ), interact with their academic journey. Currently, students face significant challenges in tracking their academic performance, understanding attendance eligibility, selecting appropriate courses and minors, and receiving timely academic guidance.

This project proposes the development of an intelligent academic management platform that consolidates scattered academic information into a single unified system. The platform integrates an AI-powered personal assistant that leverages machine learning algorithms to provide personalized academic guidance, predict examination outcomes based on continuous assessment marks, and proactively warn students about potential risks such as attendance non-eligibility.

Unlike existing student information systems that primarily serve administrative purposes, PES is designed from the student's perspective. It empowers students with real-time insights into their academic standing, provides intelligent course recommendations, enables digital medical certificate submissions and examination enrollments, and creates a feedback loop between students and the faculty administration.

The system also provides an administrative interface where faculty staff can publish examination results, update attendance records, manage course information, and analyze student feedback—creating a complete ecosystem for academic performance management.

2. Problem Statement & Objectives

The Faculty of Engineering at USJ currently lacks a centralized digital platform for students to monitor their academic performance comprehensively. The following critical problems have been identified through direct observation and student surveys:

➤ **Attendance Unawareness**

According to the Faculty Handbook (2023), students must maintain a minimum of 80% attendance in each course to be eligible for the final examination. However, there is no real-time system for students to track their attendance percentage. Students often realize they have fallen below the threshold only after it is too late, resulting in course non-eligibility. Furthermore, students who miss lectures due to medical reasons must submit medical certificates within 14 days (Faculty Handbook, Section 4.3), but there is no automated reminder system for this critical deadline.

➤ **Course Selection & Minor Confusion**

The Faculty of Engineering offers minor specializations. For example Computer Engineering department students have two minors **High performance & Data management**. Students must select specific elective courses aligned with their chosen minor. However, no centralized system clearly maps which courses belong to which minor, what prerequisites exist, or how credit requirements are distributed across semesters. Students frequently make incorrect course selections, which can delay their graduation or result in ineligibility for their desired minor.

➤ **Scattered Academic Records**

Academic records, examination results, course details, and timetables are currently dispersed across multiple systems and communication channels. Results are often shared via notice boards or informal channels. There is no single platform where students can view their complete academic history, calculate their CGPA, or analyze their performance trends across semesters.

➤ **Lack of Proactive Academic Guidance**

Currently, there is no system that proactively warns students about potential academic risks. Students rely on informal networks asking seniors or peers for guidance on course selection, examination preparation, and understanding academic regulations. This informal information sharing often leads to incomplete, outdated, or inaccurate advice, contributing to poor academic planning and preventable failures.

➤ **No Prediction or Early Warning Mechanism**

Existing systems at Sri Lankan universities do not offer predictive analytics for student performance. No mechanism exists to estimate a student's likely final grade based on mid-semester marks, continuous assessment scores, and attendance patterns. Such predictive capabilities could enable early intervention, allowing students and faculty to take corrective action before it is too late.

Aim

To design and develop a comprehensive, AI-integrated web-based platform that empowers undergraduate students with real-time academic performance tracking, proactive guidance, and intelligent insights, while providing faculty administration with efficient tools for academic management.

Specific Objectives

1. Develop a centralized student dashboard displaying complete academic history, CGPA calculations, and semester-wise performance trends.
2. Implement real-time attendance tracking with automated eligibility calculations and proactive warning notifications when attendance drops below the 80% threshold.
3. Build an AI-powered conversational assistant using NLP and Decision Tree (CART) algorithms that can answer academic queries, predict examination outcomes, and provide personalized study recommendations.
4. Create a course and minor guidance module that helps students understand module requirements, prerequisites, and elective options
5. Develop digital workflows for medical certificate submission, course enrollment, and examination registration to replace existing manual processes.
6. Implement an anonymous and non-anonymous student feedback system to enhance teaching quality and course content improvement.
7. Create an administrative panel enabling faculty staff to publish results, update attendance, manage courses, and analyze student feedback.
8. Design the system with a modular, scalable architecture that can be adapted for other faculties and universities beyond the initial deployment.

3. Literature Review & Existing Systems

This section reviews existing academic management systems both internationally and within Sri Lanka, identifies their strengths and limitations, and establishes the research gap that PES aims to fill.

3.1 International Systems

➤ Degree Works (Ellucian)

Degree Works is a degree auditing and academic advising tool used in North American universities. It tracks degree completion progress and provides 'what-if' scenarios for course selection (Ellucian, 2023). While it offers course planning capabilities, it does not provide AI-driven result predictions, real-time attendance monitoring, or a conversational AI assistant.

➤ EduTrack – Student Performance Monitoring System

EduTrack is a commercial student performance monitoring system used in several international universities. It provides GPA tracking, grade analytics, and basic performance dashboards for students and administrators. However, EduTrack does not include AI-powered academic guidance, attendance-based eligibility monitoring, or predictive analytics for examination results. It primarily functions as a retrospective reporting tool rather than a proactive student support platform (Chen & Zhang, 2021).

➤ Blackboard Analytics for Learn

Blackboard Analytics provides institutional-level reporting and student performance dashboards. It offers retention risk indicators and basic predictive models for identifying at-risk students. However, it lacks a conversational AI assistant, does not provide course-specific result predictions based on continuous assessments, and is not designed to handle institution-specific academic regulations such as attendance eligibility thresholds (Blackboard Inc., 2024).

3.2 Sri Lankan Systems

➤ Performance Evaluation Service (PES) – Faculty of Applied Sciences, USJ

The Faculty of Applied Sciences at the University of Sri Jayewardenepura operates a Performance Evaluation Service (PES) accessible at pes.science.sjp.ac.lk. This system allows students to log in and view their semester results. Developed by Avomarks Solutions, it provides basic result viewing functionalities and online course enrollment facilities. However, the system is limited to displaying published results and does not offer real-time attendance tracking, AI-powered academic guidance, predictive analytics, course selection guidance, or proactive warning systems. It functions primarily as a result publication portal rather than a comprehensive academic performance management platform (University of Sri Jayewardenepura, Faculty of Applied Sciences, 2024).

➤ University of Moratuwa – Student Information System

The University of Moratuwa operates a Student Information System (SIS) that manages student registration, examination results, and basic academic records. However, this system primarily serves administrative functions and does not provide students with real-time performance analytics, AI-powered guidance, or proactive attendance warnings (University of Moratuwa, 2024).

➤ University of Colombo – Student Management System

The University of Colombo's student management system handles enrollment, examination scheduling, and result publication. While functional for administrative purposes, it lacks personalized student dashboards, attendance tracking capabilities, course selection guidance, and any form of artificial intelligence integration (University of Colombo, 2024).

3.3 Comparison of Existing Systems

Feature	Degree Works	EduTrack	Blackboard Analytics	USJ FAS PES	UoM SIS	UoC SMS	PES (Ours)
Academic Records & Results	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Real-time Attendance Tracking	No	No	No	No	No	No	Yes
Attendance Eligibility Warnings	No	No	No	No	No	No	Yes
CGPA Calculation & Trends	Yes	Yes	Yes	Yes	Yes	No	Yes
AI-Powered Personal Assistant	No	No	No	No	No	No	Yes
Result Prediction (ML)	No	No	Yes	No	No	No	Yes
Course & Minor Guidance	Yes	No	No	No	No	No	Yes
Digital Medical Submissions	No	No	No	No	No	No	Yes
Online Exam Enrollment	No	No	No	Yes	Yes	Yes	Yes
Student Feedback System	No	No	No	No	No	No	Yes

Table 1: Feature comparison between existing systems and PES

3.4 Research Gap

As Table 1 demonstrates, no existing system either internationally or within Sri Lanka provides a comprehensive solution that combines real-time attendance tracking with eligibility warnings, AI-powered academic guidance, machine learning-based result predictions, and faculty-specific regulation handling in a single unified platform. The proposed PES addresses this gap by integrating all these features into one student-centric system specifically designed for the Sri Lankan university context, while being flexible enough to adapt to other faculties and universities.

4. Novelty & Unique Features

The novelty of the Performance Evaluation System extends far beyond simply targeting the Faculty of Engineering students as its user base. PES introduces several innovative features that are not available in any existing system used in Sri Lankan universities:

4.1 AI-Powered Conversational Academic Assistant

PES integrates a Natural Language Processing (NLP)-based AI assistant that students can interact with conversationally. Unlike static FAQ pages or manual browsing, the AI assistant can:

- Answer complex academic queries in natural language (e.g., "What courses do I need for the Data Management minor?")
- Access all system databases in real-time to provide personalized responses based on the student's actual academic records
- Provide contextual guidance based on faculty regulations from the Faculty Handbook
- Generate motivational suggestions when a student's performance shows declining trends

4.2 Machine Learning-Based Result Prediction

PES implements a Decision Tree (CART) algorithm trained on historical student performance data to predict final examination outcomes. This feature enables:

- Early identification of students at risk of failing based on mid-semester and CA marks
- Personalized study recommendations based on predicted weak areas
- Faculty-level analytics showing predicted pass/fail rates before examinations

4.3 Proactive Warning & Alert System

Unlike existing systems that are purely reactive (showing data only when requested), PES actively monitors student metrics and generates warnings:

- **Attendance Alerts:** Automatic warnings when attendance drops below 85% (pre-warning) and 80% (critical)
- **Medical Certificate Reminders:** 14-day deadline tracking with automated reminders
- **Performance Decline Alerts:** Notifications when GPA shows significant drops between semesters
- **Enrollment Deadlines:** Reminders for course enrollment and exam registration periods

4.4 Faculty-Specific Regulation Engine

PES encodes the specific academic regulations of the Faculty of Engineering (from the Faculty Handbook) into its business logic. This includes the 80% attendance rule, GPA calculation methodology (credit-weighted), minor specialization requirements, examination re-sit rules, and grading scales. This makes the system deeply integrated with the actual academic framework rather than being a generic tool.

4.5 Integrated Digital Workflows

PES digitizes currently manual processes including:

- Medical certificate submission and approval workflow
- Online course enrollment and exam registration
- Anonymous and non-anonymous course feedback submission
- Timetable management and conflict detection

4.6 Scalability & Adaptability

While initially designed for the Faculty of Engineering at USJ, the system architecture is modular and configurable. The regulation engine, grading scales, attendance thresholds, and course structures can be reconfigured for any faculty or university, making PES adaptable beyond its initial deployment scope. This is a key differentiator the novelty is not limited to the user category but lies in the system's intelligent features and architectural flexibility.

5. Dataset Overview

The system requires the following datasets to function effectively. All data will be obtained through official channels with proper permission from the Faculty administration.

Dataset	Key Fields	Purpose	Source
Student Information	student_id, reg_num, name, batch, department	Identify students and link records	Faculty Database
Course Results	student_id, course_id, mid_sem, ca_marks, final_grade	GPA/CGPA calculation, AI predictions	Examination Branch
Attendance Records	course_id, student_id, lecture_date, status	Track attendance percentages & alerts	Course Coordinators
Course Information	course_code, name, credits, semester, minor_category	Course selection & minor guidance	Faculty Handbook
Lecturer Information	lecturer_id, name, department, courses	Link courses to lecturers	Faculty Administration
Feedback Data	student_id, course_id, rating, comments, anonymous	Teaching quality improvement	Student Submissions
Historical Results	batch_year, course_id, grade_distribution	Benchmarking & AI training data	Past Exam Records

Table 2: Required datasets for PES

5.1 Data Privacy & Ethics

All student data will be handled in compliance with data protection principles. Personal information will be encrypted at rest and in transit. Role-based access control (RBAC) ensures that students can only view their own records. Administrative access is logged and auditable. Anonymous feedback ensures student privacy when providing course evaluations. Explicit consent will be obtained for any data used in AI model training.

6. System Architecture

The Performance Evaluation System follows a three-tier architecture pattern consisting of the Presentation Layer (Frontend), Application Layer (Backend & AI), and Data Layer (Databases). The system supports two primary user roles Students and Administrators each with distinct functionalities.

6.1 Architecture Description

Presentation Layer: Build with React.js and TypeScript, the frontend provides responsive, role-based user interfaces. Students access their personalized dashboard, academic records, attendance data, AI assistant, and enrollment modules. Administrators access the management panel for publishing results, updating attendance, and managing courses.

Application Layer: The backend, build with Node.js and Express.js, handles authentication (JWT-based), REST API endpoints, business logic processing, and the proactive warning engine. Supabase Edge Functions are using for serverless processing of specific tasks.

AI/ML Layer: A dedicated Python-based AI module processes natural language queries from students and runs the CART Decision Tree algorithm for result predictions. The AI module has read access to all databases to provide contextual, personalized responses.

Data Layer: PostgreSQL (via Supabase) stores all structured data including student records, course results, attendance records, course information, lecturer details, and feedback data. Row-Level Security (RLS) policies ensure data access is restricted based on user roles.

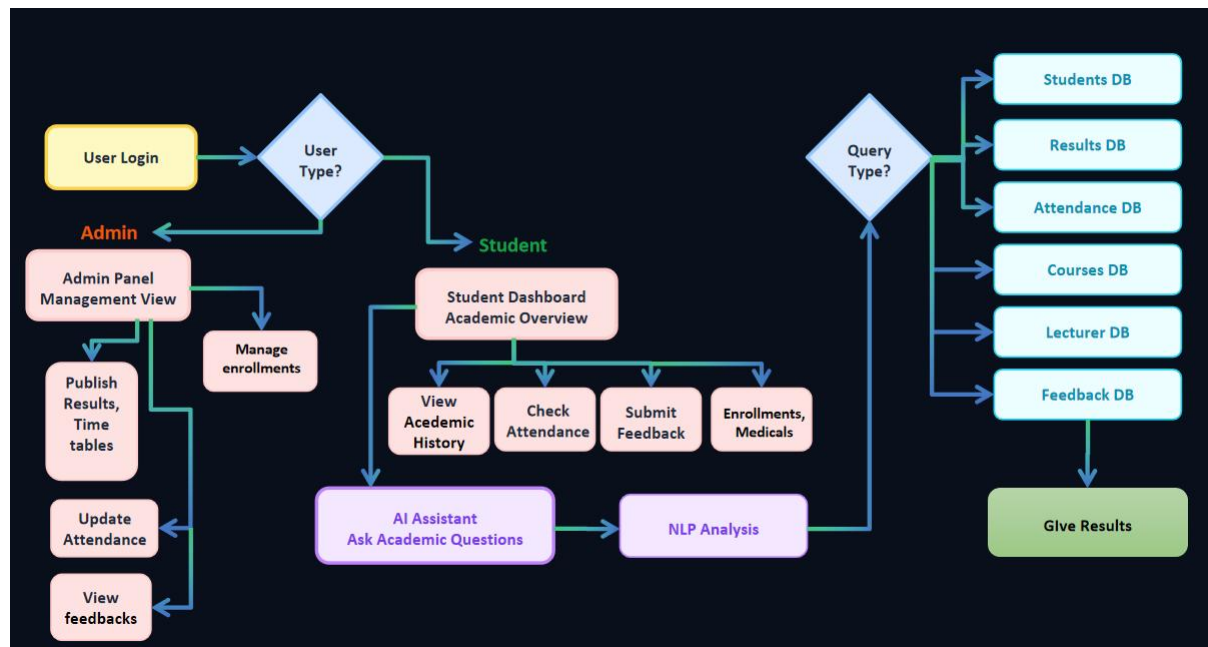


Figure 1: Proposed system architecture

7. AI Model – Training & Accuracy

The AI component of PES serves two primary functions: (1) a conversational assistant for answering academic queries, and (2) a predictive model for forecasting examination results. This section describes the approach for model training, feature selection, and accuracy evaluation.

7.1 Conversational AI Assistant

The conversational assistant utilizes a pre-trained Large Language Model (LLM) fine-tuned with faculty-specific knowledge. The system employs Retrieval-Augmented Generation (RAG) to combine the LLM's language capabilities with real-time data retrieval from the PES databases.

Training approach:

- **Knowledge Base Construction:** The Faculty Handbook, course catalogs, academic regulations, examination rules, and FAQ documents are processed and stored as vector embeddings.
- **RAG Pipeline:** When a student asks a question, the system retrieves relevant documents from the knowledge base, combines them with the student's academic context (from the database), and generates a contextual response.
- **Continuous Learning:** Student interactions are logged (with consent) to identify common queries and improve response quality over time.

7.2 Result Prediction Model

The result prediction model uses a **Classification and Regression Trees (CART)** algorithm, chosen for its interpretability and effectiveness with structured academic data.

➤ Feature Selection

The following features are used as inputs to the prediction model:

Feature	Type	Description
Mid-semester Marks	Numeric (0-100)	Mid-semester examination score
CA Marks	Numeric (0-100)	Continuous assessment total
Attendance %	Numeric (0-100)	Lecture attendance percentage
Previous GPA	Numeric (0-4.0)	Cumulative GPA up to current semester
Course Credit Weight	Numeric (1-4)	Credit value of the course
Assignment Completion Rate	Numeric (0-100)	Percentage of submitted assignments
Course Difficulty Level	Categorical	Based on historical pass rates

Table 3: Features used in the CART prediction model

➤ Training Process

- **Data Collection:** Historical academic records from the past 3-5 years (approximately 500-1000 student records per course) will be collected from the Faculty Examination Branch with official permission.
- **Data Preprocessing:** Missing values will be handled using mean imputation for numeric features and mode imputation for categorical features. Data will be normalized using Min-Max scaling.
- **Train-Test Split:** The dataset will be split into 80% training data and 20% testing data using stratified sampling to maintain class distribution.
- **Model Training:** The CART algorithm will be trained with hyperparameter tuning (max depth, min samples split, min samples leaf) using 5-fold cross-validation.
- **Output:** The model predicts the student's likely grade category: A+/A, A-/B+, B/B-, C+/C, C-/D+, or F.

➤ Accuracy Evaluation

The model's accuracy will be measured using the following metrics:

- **Overall Accuracy:** Percentage of correctly predicted grades across all categories. Target: $\geq 75\%$.
- **Precision:** For each grade category, the proportion of correct predictions out of all predictions for that category.
- **Recall:** For each grade category, the proportion of correctly identified instances out of all actual instances.
- **F1-Score:** Harmonic mean of precision and recall, providing a balanced measure. Target: ≥ 0.70 .
- **Confusion Matrix:** A detailed matrix showing the distribution of predictions vs. actual outcomes for each grade category.
- **Cross-Validation Score:** 5-fold cross-validation mean accuracy to ensure the model generalizes well and is not overfitting.

➤ Model Improvement Strategy

If the initial CART model does not achieve the target accuracy, the following strategies will be employed:

- **Ensemble Methods:** Random Forest or Gradient Boosting to combine multiple decision trees for improved accuracy.
- **Feature Engineering:** Creating derived features such as 'improvement trend' (change in marks across assignments) and 'peer comparison' (performance relative to batch average).
- **Hyperparameter Optimization:** Grid search or Bayesian optimization for finding optimal tree parameters.
- **Incremental Learning:** Retraining the model each semester with new data to improve predictions over time.

8. Tools & Technologies

Category	Technology	Purpose
Frontend Framework	React.js with TypeScript	Building interactive, type-safe user interfaces
UI Styling	Tailwind CSS + shadcn/ui	Responsive, consistent design system
Backend Runtime	Node.js with Express.js	RESTful API development and server-side logic
Database	PostgreSQL (via Supabase)	Relational data storage with RLS security
Authentication	Supabase Auth (JWT)	Secure user authentication and role-based access
File Storage	Supabase Storage	Medical certificates, documents upload
AI/ML Framework	Python, Scikit-learn	CART decision tree training and predictions
NLP Processing	LangChain + LLM API	Conversational AI assistant with RAG
Data Processing	Pandas, NumPy	Data cleaning, preprocessing, feature engineering
Version Control	Git + GitHub	Source code management and collaboration
Deployment	Vercel (Frontend), Supabase (Backend)	Cloud hosting and serverless functions
API Testing	Postman	API endpoint testing and documentation
Project Management	Notion / Trello	Task tracking and milestone management

Table 4: Technology stack for PES development

9. Implementation Plan & Timeline

Phase	Duration	Key Activities	Deliverables
Phase 1: Research & Planning	Month 1-2	Requirements gathering, literature review, system design, database schema design	SRS document, ER diagrams, wireframes
Phase 2: Foundation	Month 3-4	Database setup, user authentication, basic UI framework, API structure	Working auth system, database schema, basic frontend
Phase 3: Core Features	Month 5-6	Academic records module, attendance tracking, course enrollment, result publishing	Student dashboard, admin panel, attendance system
Phase 4: AI Integration	Month 7-8	AI assistant development, CART model training, prediction system, NLP pipeline	Working AI chatbot, prediction model with $\geq 75\%$ accuracy
Phase 5: Advanced Features	Month 9-10	Feedback system, medical submissions, warning engine, analytics dashboards	Complete feature set, proactive warning system
Phase 6: Testing & Deployment	Month 11-12	User testing, bug fixes, performance optimization, deployment, documentation	Production-ready application, user manual, source code

Table 5: Implementation timeline and deliverables

10. Expected Outcomes

Upon successful completion, the Performance Evaluation System is expected to deliver the following outcomes:

10.1 Deliverables

1. **Fully Functional Web Application:** A production-ready, responsive web platform accessible to students and administrators of the Faculty of Engineering, USJ, with secure authentication and role-based access control.
2. **AI-Powered Academic Assistant:** A conversational AI chatbot capable of answering academic queries, providing course guidance, and offering personalized study recommendations using RAG and real-time database access.
3. **ML-Based Result Prediction Model:** A trained CART decision tree model achieving $\geq 75\%$ overall accuracy in predicting student grade categories based on continuous assessment and attendance data.
4. **Proactive Alert System:** An automated warning engine that monitors student attendance and performance metrics, generating timely notifications for attendance non-eligibility risks, medical certificate deadlines, and performance declines.
5. **Administrative Dashboard:** A comprehensive management panel for faculty administrators to publish results, update attendance, manage courses and lecturers, and view aggregated analytics.
6. **Digital Workflow System:** Online modules for medical certificate submission, course enrollment, examination registration, and student feedback replacing existing manual paper-based processes.

10.2 Expected Benefits

1. **Improved Student Awareness:** Students will have real-time visibility into their attendance percentages, CGPA, and academic standing, enabling informed decision-making throughout the semester.
2. **Reduced Academic Failures:** The proactive warning system and AI-driven predictions are expected to reduce preventable course failures by alerting students early about attendance and performance risks.
3. **Enhanced Administrative Efficiency:** Digital workflows will reduce the time and effort required for result publication, attendance management, and medical certificate processing by an estimated 60-70%.
4. **Data-Driven Decision Making:** Faculty administrators will gain access to aggregated analytics and performance trends, enabling evidence-based improvements in teaching and curriculum design.
5. **Scalable Model for Sri Lankan Universities:** The modular architecture and configurable regulation engine will serve as a template that can be adapted for other faculties and universities across Sri Lanka.

11. Conclusion

This proposal presents the design and development plan for the Performance Evaluation System (PES), an intelligent academic management platform tailored for the Faculty of Engineering at the University of Sri Jayewardenepura. The system addresses critical gaps in the current academic management infrastructure, where students lack centralized access to their academic records, real-time attendance tracking, proactive academic guidance, and predictive performance insights.

Through the integration of AI-powered conversational assistance using Retrieval-Augmented Generation (RAG) and machine learning-based result prediction using the CART algorithm, PES goes beyond conventional student information systems. It transforms academic data into actionable insights, enabling students to make informed decisions about their academic journey while providing faculty administrators with efficient digital tools for academic management.

The literature review demonstrates that no existing system either internationally or within Sri Lanka provides the comprehensive combination of features that PES offers: real-time attendance monitoring with eligibility warnings, AI-powered academic guidance, machine learning predictions, faculty-specific regulation handling, and integrated digital workflows, all within a single unified platform.

The feasibility of this project is supported by the well-defined technology stack (React.js, Node.js, PostgreSQL, Python, Scikit-learn), the availability of required datasets through official faculty channels, and a structured 12-month implementation timeline with clear milestones and deliverables. The modular architecture ensures that the system can be extended and adapted for other faculties and universities, maximizing its long-term impact.

In summary, PES represents a significant step forward in leveraging data management and artificial intelligence to improve the undergraduate academic experience in Sri Lankan universities. By addressing the identified problems with innovative, student-centric solutions, this project has the potential to serve as a model for academic performance management systems across the higher education sector.

12. References

1. Faculty of Engineering, University of Sri Jayewardenepura (2023). Student Guide – Faculty Handbook 2023. University of Sri Jayewardenepura. Available at: <https://www.sjp.ac.lk/engineering/>
2. Faculty of Applied Science, University of Sri Jayewardenepura (2024). Student Information System. University of Sri Jayewardenepura. Available at: <https://pes.science.sjp.ac.lk/student/>
3. Ellucian (2023). Degree Works: Academic Advising and Degree Audit. Available at: <https://www.ellucian.com/solutions/ellucian-degree-works>
4. University of Moratuwa (2024). Student Information System. University of Moratuwa. Available at: <https://uom.lk/>
5. University of Colombo (2024). Student Management System. University of Colombo. Available at: <https://cmb.ac.lk/>
6. Breiman, L., Friedman, J., Stone, C.J. and Olshen, R.A. (1984). Classification and Regression Trees. CRC Press.
7. Pedregosa, F. et al. (2011). 'Scikit-learn: Machine Learning in Python', Journal of Machine Learning Research, 12, pp. 2825-2830.
8. Lewis, P. et al. (2020). 'Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks', Advances in Neural Information Processing Systems, 33, pp. 9459-9474.
9. Romero, C. and Ventura, S. (2010). 'Educational Data Mining: A Review of the State of the Art', IEEE Transactions on Systems, Man, and Cybernetics, 40(6), pp. 601-618.
10. Baker, R.S. and Inventado, P.S. (2014). 'Educational Data Mining and Learning Analytics', in Learning Analytics. Springer, New York, pp. 61-75.
11. Supabase Inc. (2024). Supabase Documentation: PostgreSQL, Authentication, and Edge Functions. Available at: <https://supabase.com/docs>